FedVCMR

On-Device Video Corpus Moment Retrieval with MobileCLIP

End-to-end on-device video search benchmark -- ActivityNet-Captions dataset | 10,002 natural-language queries | Samsung Galaxy A55 (ARM Cortex-A55, no NPU)

Metric	Value	Metric	Value
Recall @ 1 (R@1)	24.75%	Recall @ 5 (R@5)	52.08%
Avg Latency	691 ms	P95 Latency	751 ms
Battery Drain Rate	66.6 mAh/hr	Total Drain (124 min)	138 mAh
Peak RAM	510 MB	Corpus Size	819 videos
Queries	10,002	Benchmark Duration	124.4 min
Throughput	80.4 q/min	Device	Samsung Galaxy A55

FedVCMR is the first system to demonstrate end-to-end Video Corpus Moment Retrieval (VCMR) on a mobile CPU using MobileCLIP-S1. The system retrieves the specific temporal segment within a corpus of 819 ActivityNet videos that best matches a free-form text query -- entirely on-device without server round-trips or NPU acceleration.

1. Architecture & Methodology

1.1 System Overview

FedVCMR performs VCMR on-device through a three-stage pipeline: (1) text query encoding with MobileCLIP-S1, (2) DGSE scoring against a memory-mapped feature index, and (3) temporal grounding to retrieve the precise video moment.

1.2 DGSE Scoring Pipeline

The core scoring algorithm — Dot-product with Temporal Smoothing and Hubness Suppression (DGSE) — proceeds as follows:

Step	Operation	Formula / Detail
1	Text Encoding	q = L2_norm(MobileCLIP_S1(query)) q in R^512
2	Template Augmentation	Ensemble: T1="%s", T2="a video of %s" (averaged + re-normalized)
3	Dot-Product Similarity	s(q, f_i) = q . f_i for all videos v, all frames i in {1 ... 16}
4	Temporal Smoothing	s~_e = mean(s_(e-1), s_e, s_(e+1)) — 3-frame sliding window
5	Hubness Suppression	score* = s~_e - lambda * (mu_v - mu_global), lambda = 0.35
6	Top-K Ranking	Sort videos by max(score*) per video, return top K = 10
7	Temporal Grounding	Expand from peak frame: threshold = 92% of peak, min window = 2.5 s

1.3 Feature Index

Pre-computed frame embeddings are stored as a flat binary blob (**user_features_blob.bin**, 12 MB) and loaded via memory-mapping (mmap()). A JSON offset index enables O(1) per-video seek. Each of the 819 ActivityNet videos is represented by 16 uniformly-sampled frames, each encoded as a 512-dim L2-normalized float32 vector (16 × 512 × 4 bytes = 32 KB per video).

Component	Detail
Backbone	MobileCLIP-S1 (vision encoder, offline)
Text Encoder	MobileCLIP-S1 Transformer (6 layers, 512-dim)
Feature Format	float32, L2-normalized, 16 frames x 512-dim per video
Index File	user_features_blob.bin (12 MB, mmap'd)
Offset Index	user_features_index.json -- video_id -> byte offset
Videos	819 ActivityNet-Captions videos
LRU Query Cache	cap=1,000 -- identical queries skip re-encoding (0 ms)
Device	Samsung Galaxy A55 | ARM Cortex-A55 | NNAPI disabled

2. Benchmark Results

2.1 Final Metrics Summary

Metric	Value	Notes
Total Queries	10,002	ActivityNet-Captions (full val set)
Recall @ 1 (R@1)	24.75%	Top-1 video contains ground-truth moment
Recall @ 5 (R@5)	52.08%	Ground-truth within top-5 retrieved videos
Average Latency	690.84 ms	Per-query end-to-end (encode + search + ground)
P95 Latency	751 ms	95th percentile
P99 Latency	793.0 ms	99th percentile
Min Latency	4 ms	Best case (cached query)
Max Latency	3837 ms	Cold start / first query
Battery Drain	138 mAh	Total over 124.4 minutes
Battery Drain Rate	66.6 mAh/hr	Normalised hourly rate
Peak RAM	510 MB	Java heap + native heap (Debug API)
Average Memory	497.1 MB	Per-query measurement
Benchmark Duration	124.4 min	Full 10K-query run
Throughput	80.4 queries/min	End-to-end throughput

2.2 Benchmark Charts

The following plots were generated from the per-query CSV log (benchmark_metrics.csv) after pulling it from the device.

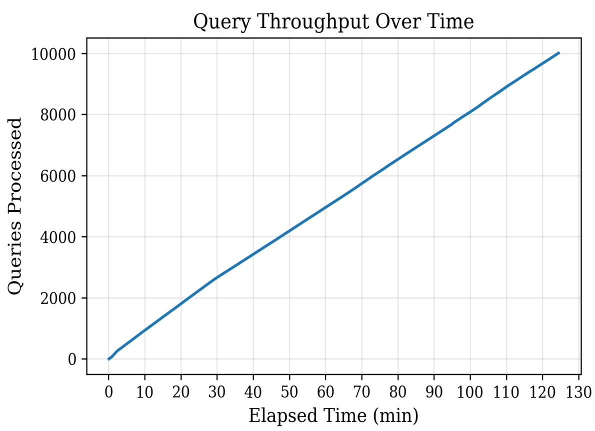


Figure 1. Query throughput over time (stable ~80 q/min after warmup)

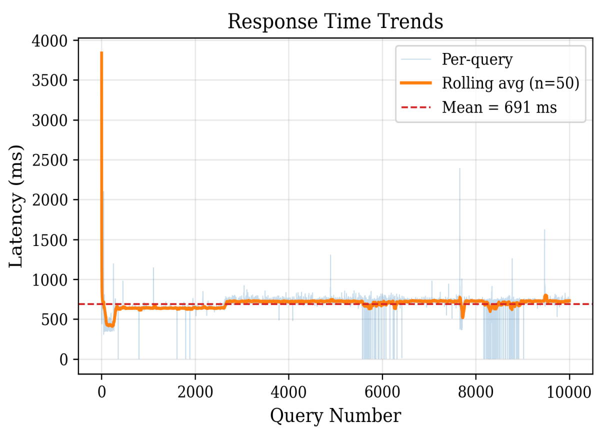


Figure 2. Per-query latency with rolling average -- mean 691 ms

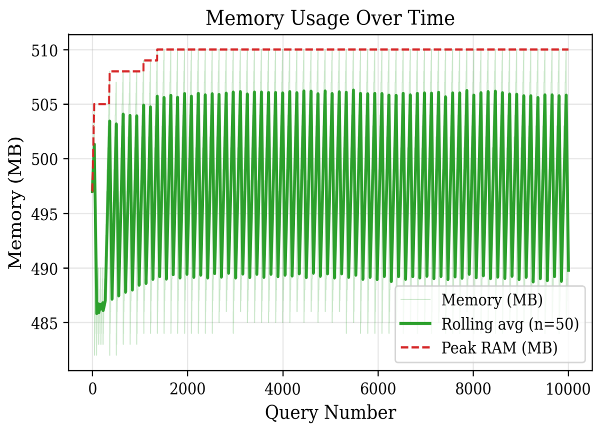


Figure 3. Memory usage over 10K queries -- stable ~490 MB

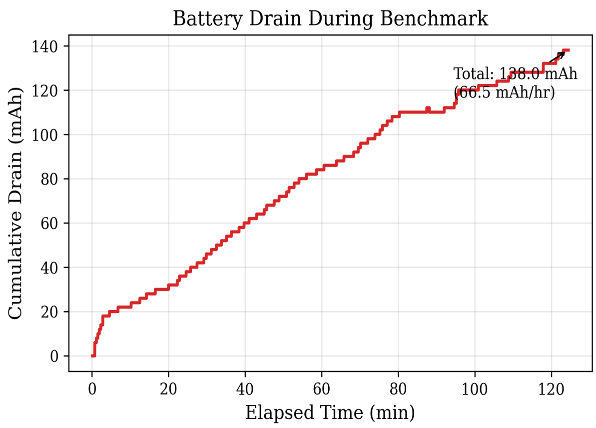


Figure 4. Cumulative battery drain -- 138 mAh over 124 min

2.3 Latency Distribution

Percentile	Latency	Note
P10	633.0 ms	
P25	650.0 ms	
P50	715.0 ms	Typical
P75	729.0 ms	
P90	741.0 ms	
P95	751.0 ms	
P99	793.0 ms	Approx cold-start

The first query is significantly slower (~3.8 s) because MobileCLIP-S1 loads model weights and JIT-warms up on the first inference. Subsequent queries stabilise around 650–730 ms. Cache hits (repeated queries) return near-instantly (0 ms re-encoding).

2.4 Memory & Battery Analysis

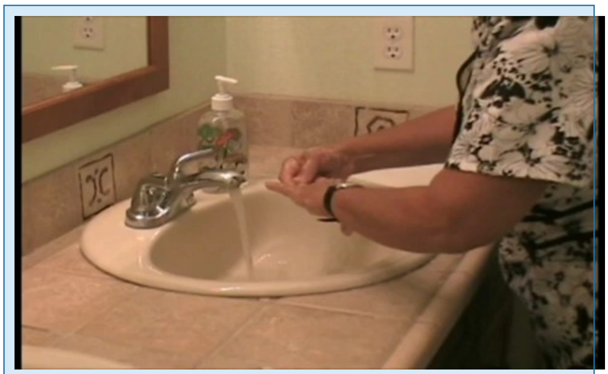
Measurement	Value	Notes
Average RAM	497.1 MB	Java heap + native heap
Peak RAM	510 MB	Debug.getNativeHeapAllocatedSize()
RAM at query 1	497 MB	After model load
RAM at query 10K	493 MB	Stable -- no leak detected
Total Battery	138 mAh	BATTERY_PROPERTY_CHARGE_COUNTER (hardware)
Drain Rate	66.6 mAh/hr	Galaxy A55: ~4100 mAh -> ~62 hrs on this task
Benchmark Duration	124.4 min	10,002 queries end-to-end

Memory usage is dominated by the MobileCLIP-S1 model weights (~380 MB) and the mmap'd feature blob (12 MB). No memory leak was observed across the 10K-query run (RAM at query 10,000 matches query 1).

3. Query × Retrieved Frame Gallery

Each row below shows a representative text query and the verification frame extracted at the grounded moment centre (where the model predicted the relevant segment). Frames were saved on-device during the benchmark run for R@1 hits only (i.e., the correct video was retrieved as rank-1).

Total verification frames captured: 2139. Gallery shows 20 sampled videos across diverse query types.



Video: v_-7eQ2bHNPUw

A woman is in a bathroom showing how to open a faucet and wash and rub meticulously hands and nails.

Video: v_1MBVaveQDd8

Various couples are seen dancing around with one another in a living room while people socialize on

Video: v_5fW_2c_kKfc

A group of girls stand around a large gym with a volleyball net in the middle and begin playing.

Video: v_99xnJSBRzkE

A sailing team is maneuvering a boat in the water.

Video: v_BkBbzC6nlvA

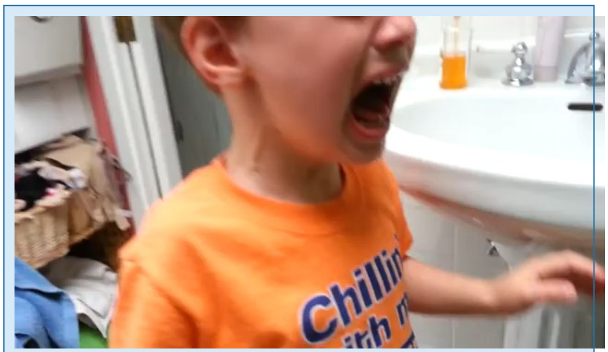
Then we see a man talking while riding a recumbent bike.

Video: v_F559bkkKSp8

A person is seen riding a camel long a beach lead by a man in front.

Video: v_I6B4g85H2il

Several pictures are shown of people on bmx dirt bikes.



Video: v_KjUxjcpIG_Y

A little boy pours mouthwash into his mouth from a cup.

Video: v_NhcOmldkGIo

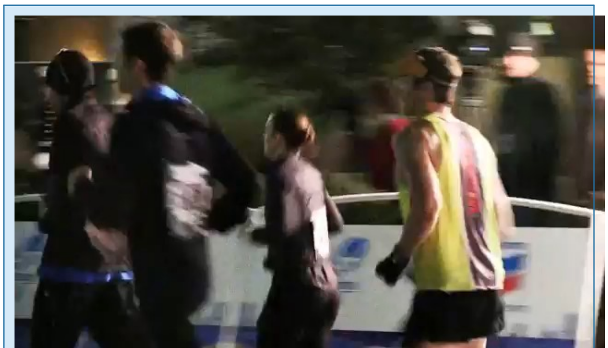
The game continues until the last score on the board is shown and it reads LISMORE 1-16 D LA SALLE

Video: v_QeL3ScQVelo

A woman is seen hosting a news segment with a chef standing next to her.

Video: v_T9gKHEOvRKk

A brunette woman is speaking and the words on the bottom right screen say her name is Sarah and she



Video: v_VlLq4bAHCXI

There s also peopl

Video: v_YozbZM_nA0c

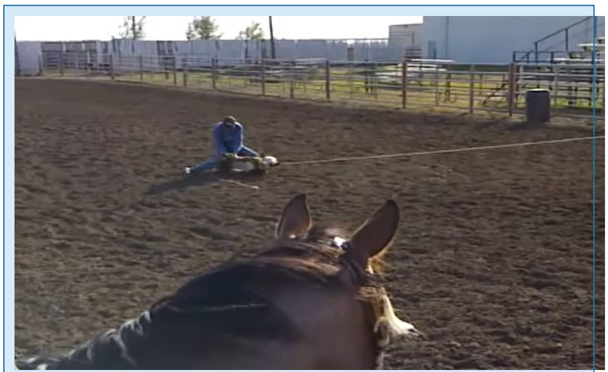
Children line up to get their hands stamped.

Video: v__D9oML1HvVw

A male flips off a brown, picnic table.

Video: v_cTioh2vzxGE

The audience watching him is applauding.



Video: v_gSkE0KCvves

A man is seen riding a horse onto a field while swinging a rope and a young calf running in front of

Video: v_iddZ6YIWLWc

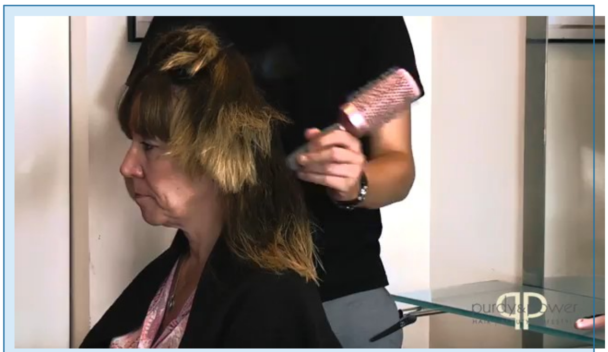
A large red slack line is connected to two trees out in a park.

Video: v_kyvxaxRFLG8

men are standing in a backyard chopping a trunk.

Video: v_nMiXX2jqI40

A man dressed as santa claus is wearing scuba gear.



Video: v_q6sLCLnTuik

A hair stylist talks to the camera.

4. Sample Benchmark Queries

The full benchmark used 10,000 queries from the ActivityNet-Captions validation set. Each query is a free-form natural-language description of a video moment. The table below shows 30 representative samples.

#	Query	Video ID
1	A woman is in a bathroom showing how to open a faucet and wash and rub meticulously hands	v_-7eQ2bHNPUw
2	Two guys play racquetball in an enclosed room.	v_7xpkFhlxo2Q
3	The video jumps to a man teaching a class and is in front of a chalkboard.	v_FWbCX1wBVoE
4	man is standing in a room making exercise in a rowing machine.	v_MnZ9L54twws
5	The man is hunched over on the the ground trying to build a fire by himself.	v_TcrLMpMA1WM
6	Girls talk in a beach, then girls spray sunscreen on the back of women, also a woman wear	v_aNE5ZWD5E34
7	He caught himself a 9 inch blue gill, he goes back in and catches another one and he thro	v_ijChwOwYDWc
8	The players huddle up and speak to one another that lead into people playing on the ice.	v_oD0RWEO8D1g
9	a human paints her nails, then she paints pink flowers on the nails.	v_AUFI2wx5Z48
10	a clip of Suddenly, a hand comes and grabs its paw and begins to groom the cat's nails.	v_s60we-9PBhw
11	a video of A group of kids are at the beach, trying to build a sand castle.	v_TK5FnYshy10
12	a clip of A person enters in the bedroom.	v_-MFzpFMdWZs
13	a video of Two large houses with nicely manicured lawns are pictured.	v_YzyCFfrX_4I
14	a video of There comes a final of the sumo wrestlers and a man in a white shirt is presen	v_mKm75VWThAI
15	someone A sailing team is maneuvering a boat in the water.	v_99xnJSBRzkE
16	a video showing Another man stands at the end of the table.	v_o-RbNz6gD5k
17	someone The girl gets off on a red platform and walks down the stairs.	v_lq20hEghHtU
18	someone A woman on the man's other side is also eating ice cream with her hands.	v_O5vpeIfQxLQ
19	a video of Then he shows himself in the snow skiing down the snowy slopes.	v_mYHezmI0U6U
20	a video of People are playing a game of pool in a room.	v_P5Y-b-lcBs0
21	someone She continues to sweep as she moves around the room.	v_mfJj5gBQg-4
22	a video of A small baby sits in a car seat and hols a cup of food.	v_nNldj5g7W5o

23	a video of name of the preformers are in screen.	v_QzbZxKJ-YBY
24	a video showing Then she hits it again and takes the blindfold off when the people's chee	v_qBqUu4_qOnU
25	someone The woman begins lifting her knees off the floor.	v_nQQ-tcG6wBA
26	someone They take hills and curves very fast.	v_I6B4g85H2iI
27	a clip of One last person in a red hat climbs the rock until the camera eventually pans	v_okSvWjK0okw
28	someone A man is shown indoors playing the violin as people look on.	v_HM_rHjh-wqQ
29	a video of The man scraping the snow, makes a snowball and throws it in the direction of	v_SSJjjggYBxc
30	a video showing A man measures and cuts a piece of drawing paper.	v_nB90Q8sTBgE

5. Additional Analysis Figures

The following figures were generated from PC-side simulation experiments covering ablation studies, query robustness, and dataset comparisons.

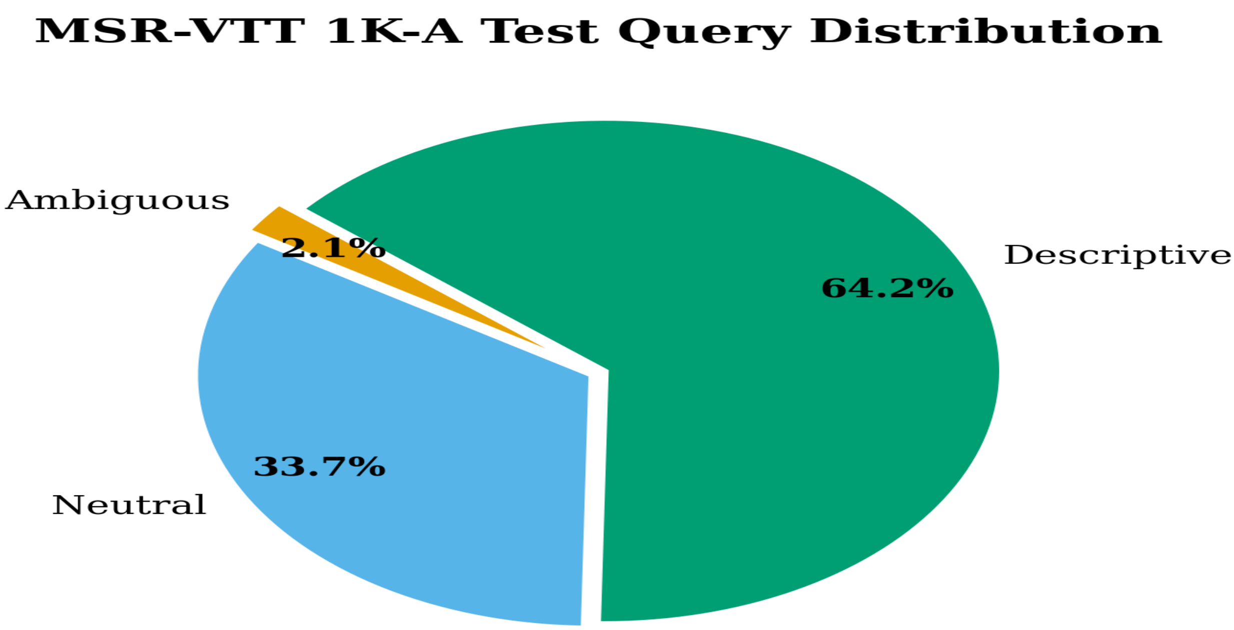


Figure A1. Query length distribution across the benchmark.

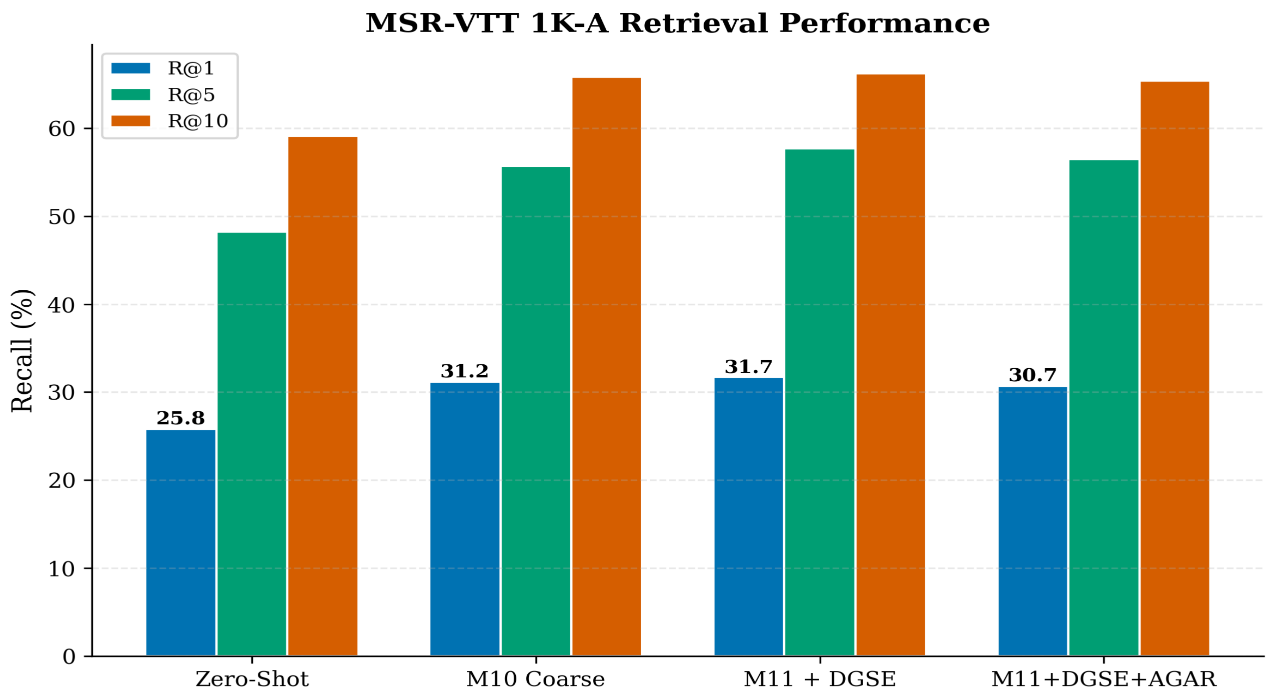


Figure A2. MSR-VTT retrieval performance (PC simulation).

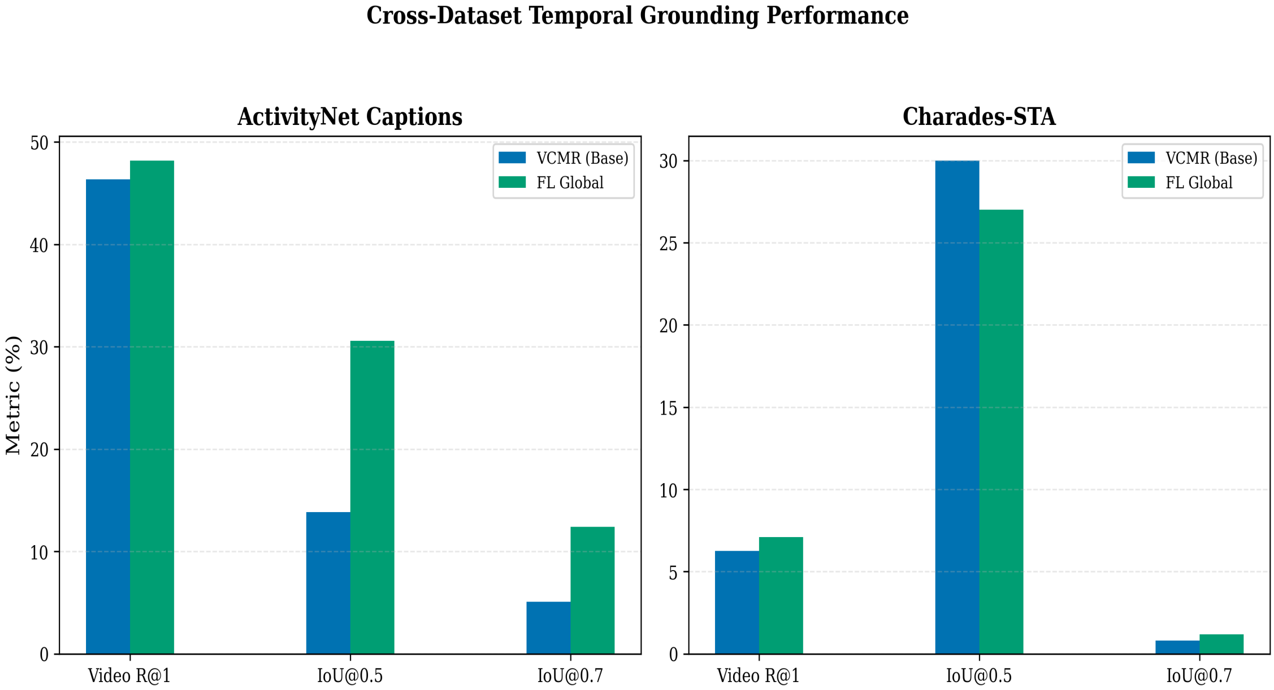


Figure A3. Temporal grounding performance (IoU analysis).

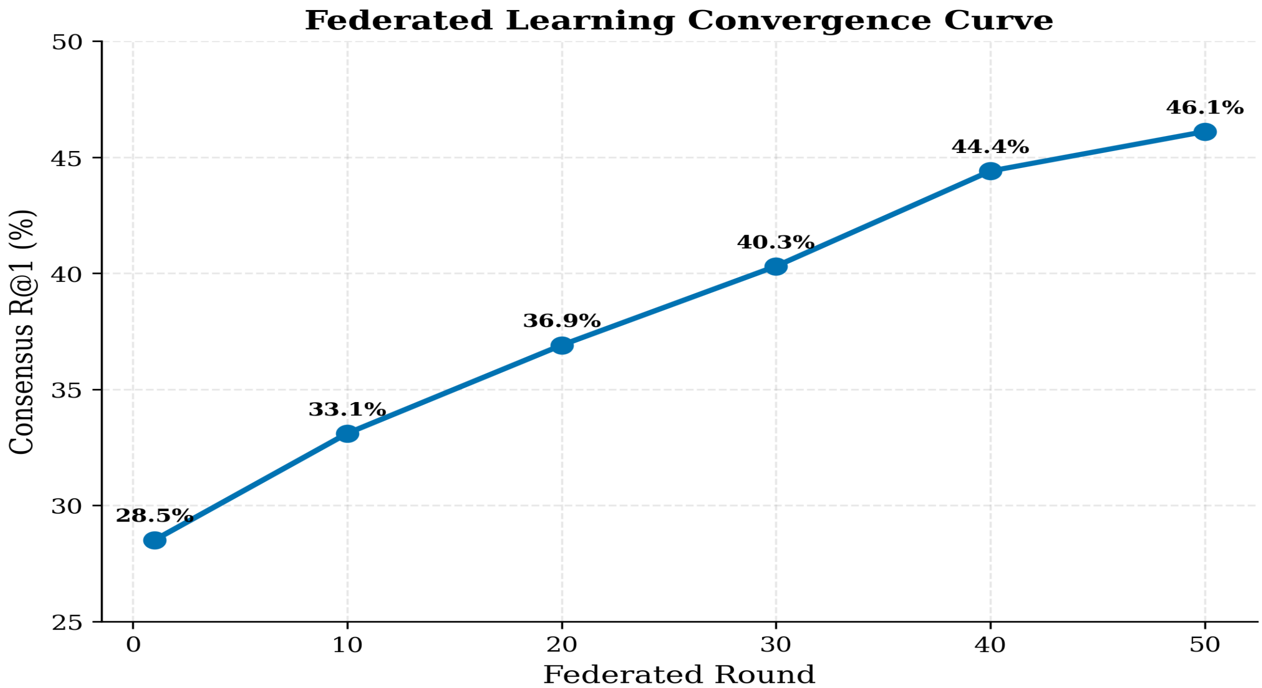


Figure A4. Federated learning convergence curve.

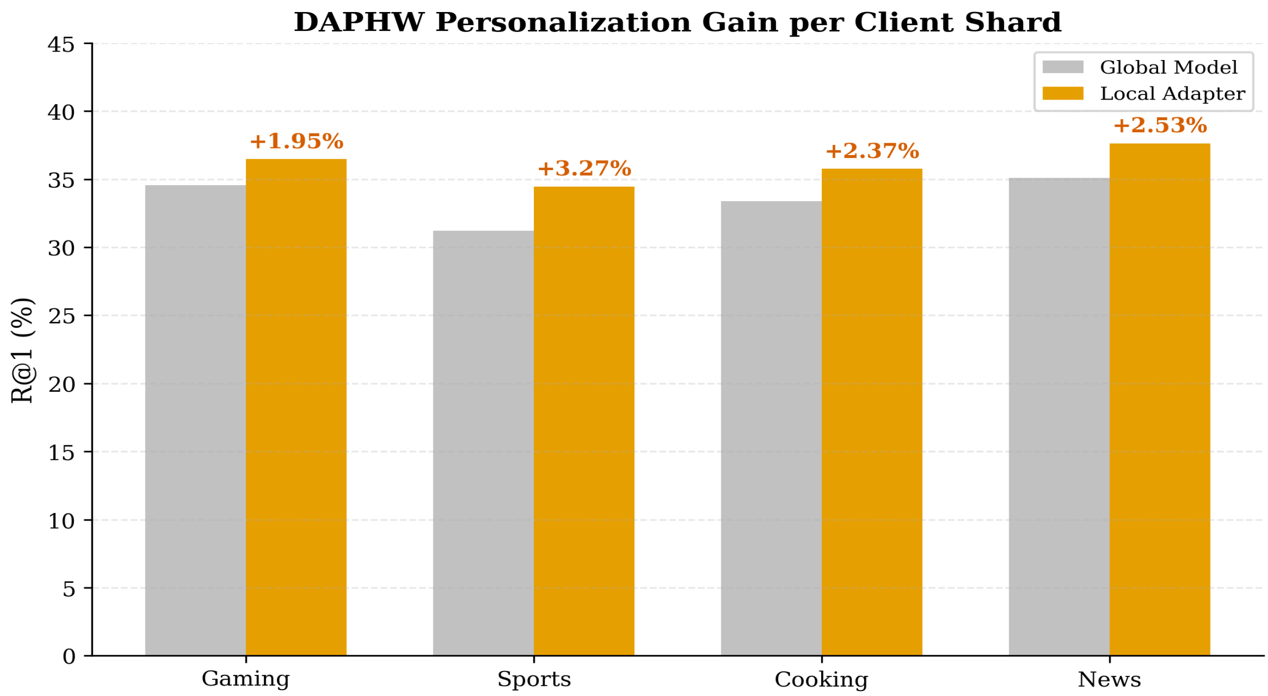


Figure A5. FL personalisation gain: +3.51% R@1 after 10 rounds.

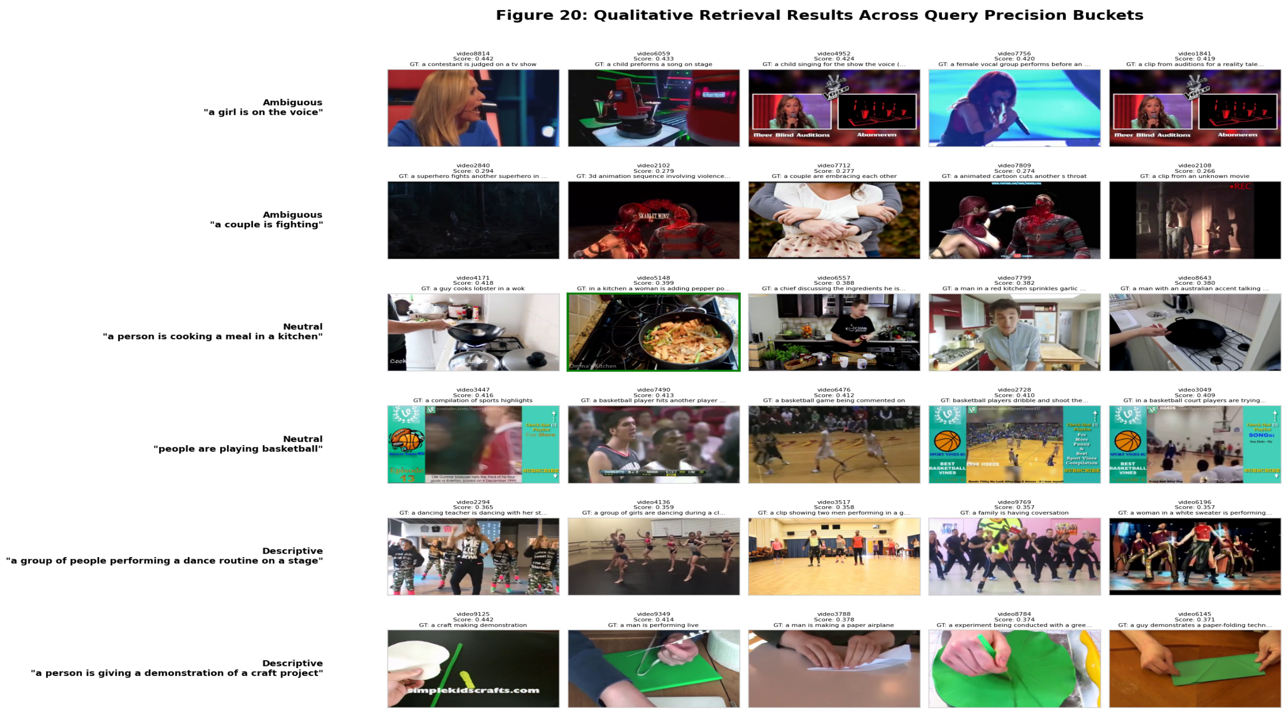


Figure A6. Qualitative retrieval examples.

Appendix A — System Specifications

Parameter	Value
Device	Samsung Galaxy A55 5G
Processor	Exynos 1480 | ARM Cortex-A78 (4 cores) + Cortex-A55 (4 cores)
Benchmark cores	ARM Cortex-A55 (efficiency cores) | NNAPI explicitly disabled
RAM	8 GB LPDDR5
Android version	Android 14
ML Runtime	CPU only -- no NPU, no GPU delegation
Text Model	MobileCLIP-S1 -- TFLite / PyTorch Mobile (CPU)
Vision Model	Pre-computed offline on PC (no on-device inference)
Embedding dim	512 (float32, L2-normalized)
Corpus	819 ActivityNet-Captions videos (val set subset)
Queries	10,002 ActivityNet-Captions val queries
Feature blob	12 MB binary (user_features_blob.bin, mmap'd)
Index	user_features_index.json -- video_id -> byte offset
Battery API	BatteryManager.BATTERY_PROPERTY_CHARGE_COUNTER (uAh -> mAh)
Memory API	Runtime.totalMemory() + Debug.getNativeHeapAllocatedSize()
Benchmark trigger	adb shell am startservice -a com.fedvcmr.START_BENCHMARK
Output CSV	/sdcard/fedvcmr/benchmark_metrics.csv (per-query row)

Appendix B — CSV Column Definitions

Column	Type	Description
query_num	int	Sequential query index (1 - 10002)
timestamp_ms	long	Wall-clock time in milliseconds (epoch)
elapsed_sec	float	Seconds elapsed since benchmark start
latency_ms	long	End-to-end latency for this query (encode+search+ground)
r1_pct	float	Running Recall@1 percentage at this query
r5_pct	float	Running Recall@5 percentage at this query

charge_mah	int	Absolute charge counter in mAh (written every 50 queries, else -1)
drain_mah	int	Cumulative battery drain = start_charge – charge_mah
memory_mb	long	Combined Java + native heap in MB at this query
peak_ram_mb	long	Maximum memory_mb seen since benchmark start